LifeMate: Agent Benchmark and Ablation

Train split · 1,545 samples · LFM3-1.7B and GIPformer 1.5-68M

Latest run Sandbox 3.3: llama.cpp Thinking: on ASR: FP32

Previous runs Transformers Thinking off ASR FP16

Table 1: Cumulative ablation results. Values are percentages except event counts. Best values are bold.

	Contest	State		Behavior		Exploratory
Configuration	Contest Score ↑	SSR ↑	USMR ↓	SHR (Rec./Err.) ↑	MI/Abs F1 ↑	DDR ↓
Baseline	6.83	26.57	19.22	20.77 (279/1,343)	14.23	65.62 (33/96)
+ Enhanced Tool Descriptions	**7.72**	27.30	16.05	23.64 (265/1,121)	14.65	55.00 (36/80)
+ Enhanced System Prompt	7.66	27.09	**15.08**	23.83 (345/1,448)	14.88	11.63 (38/43)
+ Enhanced Sandbox Harness	7.64	26.85	16.31	**27.44** (309/1,126)	**14.93**	17.78 (37/45)
+ Sandbox 3.1 Runtime Checks	6.96	26.61	20.71	22.30 (188/843)	11.70	20.34 (47/59)
+ Sandbox 3.2 Runtime Checks	6.66	26.03	15.47	17.81 (135/758)	12.42	**8.33** (33/36)
+ Sandbox 3.3*	6.54	**30.37**	24.92	18.08 (232/1,283)	9.54	38.93 (91/149)

Contest Score approximates official scoring and may differ due to whitespace tokenization. SSR is sample-mean state recall. USMR is sample-level unintended mutation rate. SHR is recovered over total error events. MI/Abs F1 pools missing-info and abstention F1. DDR uses 1,167 paired state-changing samples; parenthetical counts in its column are full/target matches.

Table 2: Outcome and execution diagnostics on 1,545 samples, count (share). Rows may overlap. Best values are bold.

Category	Diagnostic	Baseline	+ Tool Desc.	+ Sys. Prompt	+ Sandbox	+ Sandbox 3.1	+ Sandbox 3.2	+ Sandbox 3.3*
Outcome	Final state match	367 (23.75%)	386 (24.98%)	389 (25.18%)	380 (24.60%)	371 (24.01%)	373 (24.14%)	**399 (25.83%)**
	Final state mismatch	1,178 (76.25%)	1,159 (75.02%)	1,156 (74.82%)	1,165 (75.40%)	1,174 (75.99%)	1,172 (75.86%)	**1,146 (74.17%)**
	Exact ordered tool calls	87 (5.63%)	99 (6.41%)	**100 (6.47%)**	**100 (6.47%)**	90 (5.83%)	87 (5.63%)	60 (3.88%)
	State match, gold has calls	270 (17.48%)	282 (18.25%)	286 (18.51%)	278 (17.99%)	273 (17.67%)	272 (17.61%)	**308 (19.94%)**
	State match, no calls	215 (13.92%)	237 (15.34%)	226 (14.63%)	226 (14.63%)	203 (13.14%)	211 (13.66%)	52 (3.37%)
Execution	No submitted tool calls	714 (46.21%)	755 (48.87%)	709 (45.89%)	709 (45.89%)	673 (43.56%)	701 (45.37%)	**124 (8.03%)**
	No calls, gold has calls	639 (41.36%)	675 (43.69%)	630 (40.78%)	630 (40.78%)	609 (39.42%)	636 (41.17%)	**100 (6.47%)**
	Reached 20-turn limit	95 (6.15%)	70 (4.53%)	89 (5.76%)	85 (5.50%)	62 (4.01%)	56 (3.62%)	**0 (0.00%)**
	Repeated tool calls (attempted)	161 (10.42%)	137 (8.87%)	194 (12.56%)	183 (11.84%)	196 (12.69%)	148 (9.58%)	**30 (1.94%)**
	Repeated tool calls (executed)	161 (10.42%)	137 (8.87%)	194 (12.56%)	182 (11.78%)	41 (2.65%)	27 (1.75%)	**17 (1.10%)**
	Output parsing failure	12 (0.78%)	6 (0.39%)	4 (0.26%)	**0 (0.00%)**	**0 (0.00%)**	**0 (0.00%)**	**0 (0.00%)**
	Hit context limit	**3 (0.19%)**	8 (0.52%)	9 (0.58%)	7 (0.45%)	6 (0.39%)	7 (0.45%)	23 (1.49%)

All runs use a 16,384-token context with no per-turn output cap. * Sandbox 3.3 uses llama.cpp, thinking on and ASR FP32. Its ASR differs from Sandbox 3.2 on 45 of 5,036 clips, so this row is not an isolated Sandbox ablation. Attempted repeats include precheck blocks; executed repeats reached ToolEnv.

Table 3: Language-model token usage across ablation stages.

Token measure	Configuration	Mean	Median	P90	P95	Maximum
First-turn input / sample	Baseline	4,505	4,374	6,400	6,954	8,487
	+ Tool Desc.	5,229	5,100	7,411	8,056	9,622
	+ Sys. Prompt	5,510	5,381	7,692	8,337	9,903
	+ Sandbox	5,510	5,381	7,692	8,337	9,903
	+ Sandbox 3.1	5,661	5,532	7,843	8,488	10,054
	+ Sandbox 3.2	5,583	5,454	7,765	8,410	9,976
	+ Sandbox 3.3*	5,582	5,453	7,764	8,409	9,973
Input summed over turns / sample	Baseline	26,125	7,021	85,876	140,433	258,719
	+ Tool Desc.	27,460	7,431	91,582	150,050	255,237
	+ Sys. Prompt	31,639	8,723	108,777	170,480	273,116
	+ Sandbox	30,498	8,723	99,937	155,400	286,976
	+ Sandbox 3.1	33,987	14,960	96,226	151,753	249,550
	+ Sandbox 3.2	26,677	10,097	71,621	137,776	279,365
	+ Sandbox 3.3*	35,579	25,563	77,581	101,897	183,111
Output / turn	Baseline	284	67	814	1,425	12,097
	+ Tool Desc.	302	93	855	1,464	6,555
	+ Sys. Prompt	285	52	921	1,478	13,077
	+ Sandbox	303	67	968	1,562	13,077
	+ Sandbox 3.1	290	97	849	1,424	9,479
	+ Sandbox 3.2	303	98	902	1,518	10,460
	+ Sandbox 3.3*	838	564	1,993	2,582	13,005
Output summed over turns / sample	Baseline	1,115	370	3,308	4,583	12,097
	+ Tool Desc.	1,084	328	3,359	4,664	8,634
	+ Sys. Prompt	1,138	359	3,385	4,460	13,077
	+ Sandbox	1,173	356	3,388	4,553	13,077
	+ Sandbox 3.1	1,307	568	3,613	4,831	9,479
	+ Sandbox 3.2	1,061	366	3,168	4,201	10,460
	+ Sandbox 3.3*	3,720	3,564	6,091	6,856	13,819

A turn is one generation. Input includes prompt and history, which are counted again across turns. Percentiles use index $\lfloor p(n-1) \rfloor$. Output includes all generated text.